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24 March 1983
Superseding
MIL-R-25170A(USAF)
24 March 1958

MILITARY SPECIFICATION

RELEASE, AUTOMATIC, AIRCRAFT SAFETY LAP BELT

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for a gas-pressure-operated automatic-release mechanism for use with the automatic-opening safety lap belt and shoulder harness.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-636	Box, Shipping, Fiberboard

MILITARY

MIL-P-116	Preservation, Methods of

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
* MIL-STD-810	Environmental Test Methods

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio ALC/MMEDO, Kelly AFB, Texas 78241 by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1680

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MIL-STD-1186 Cushioning, Anchoring, Bracing, Blocking and Waterproofing with Appropriate Test Methods

DRAWINGS

U.S. AIR FORCE

55C502 Release, Automatic, Aircraft Safety Lap Belt
55B600 Hose-Automatic Lap Belt, Actuating, Assembly of

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.1.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article test. This specification makes provisions for first article testing.

3.2 Design and construction. The automatic releases shall be constructed in accordance with Air Force Drawing 55C502.

3.3 Government-furnished property. When provided for in the contract or purchase order, the following items will be furnished by the Government to the contractor upon his request as required.

Initiator, Delay, M4, Part No. 94-S-7 or Initiatar, Delay, M12, Part No. 94-3-9.

Cartridge, Delay, M46, Part No. BLX-93-I or Cartridge, Delay, M71, Part No. F8493162.

The quantities of Government-furnished property are as follows:

One delay initiator, loaded and staked, and two additional delay cartridges required per lot of automatic releases submitted for acceptance. See paragraph 6.3.

3.4 Identification of product. The physical item shall be marked for identification in accordance with Standard MIL-STD-130.

* 3.5 Workmanship. Because of the emergency nature of the equipment, it is required that all details of workmanship be of the highest quality to insure proper functioning under the conditions to which the safety belts may be subjected. Unsatisfactory workmanship items such as Loose, cocked, or inadequately headed rivets, distorted, loose, or inadequately secured bushings and pins, loose, broken or incomplete stitching; and rough, malformed, misaligned

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or improperly machined fittings shall be considered as defects. All parts shall be clean and free from dirt, sand, metal chips, or foreign matter while being assembled.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of tests. The inspection and testing of releases shall be classified as follows:

a. Acceptance tests

* b. First article tests

4.2 Acceptance tests. Acceptance tests shall consist of:

a. Individual tests

b. Sampling plan and tests

4.2.1 Individual tests. Each automatic release shall be subjected to the following tests as described under 4.4, Test Methods:

a. Examination of product

b. Functional tests

4.2.2 Sampling plan and tests.

4.2.2.1 Lot. The lot definition, formation, and size shall be in accordance with MIL-STD-105.

4.2.2.1.1 MIL-STD-105. When MIL-STD-105 specifies an action by the Government, it shall, at the option of the Government, be performed either by the Government or by the contractor under the supervision of the Government inspector.

4.2.2.2 Sampling tests. Two automatic releases shall be selected at random from each lot of 200 or fraction thereof on the order. One of these releases shall be utilized for each of the following tests.

a. Low temperature automatic operation

b. Ultimate load

4.3 Test conditions. Unless otherwise specified, all tests required by this specification shall be made under the normal atmospheric conditions existing at the time and location of the tests.

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4.4 Test methods

4.4.1 Examination of product. Each automatic release shall be inspected for conformance to this specification with respect to workmanship, materials, dimensions, and markings.

* 4.4.2 Environmental tests. One automatic release shall be subjected to the following environmental tests of MIL-STD-810. At the conclusion of all environmental tests, an initiator and a 48-inch length of pressure transmission hose (P/N 55B600-48) shall be attached to the release. The following loads shall then be applied to the belt as shown in figure I: Load A, 1000 pounds, loads b and c, 0 pounds. The initiator shall then be actuated. The connecting link shall readily separate from the release.

Test	Procedure	Time
Humidity	I	240 hours
Salt Spray	I	50 hours
Sand and Dust	I	12 hours

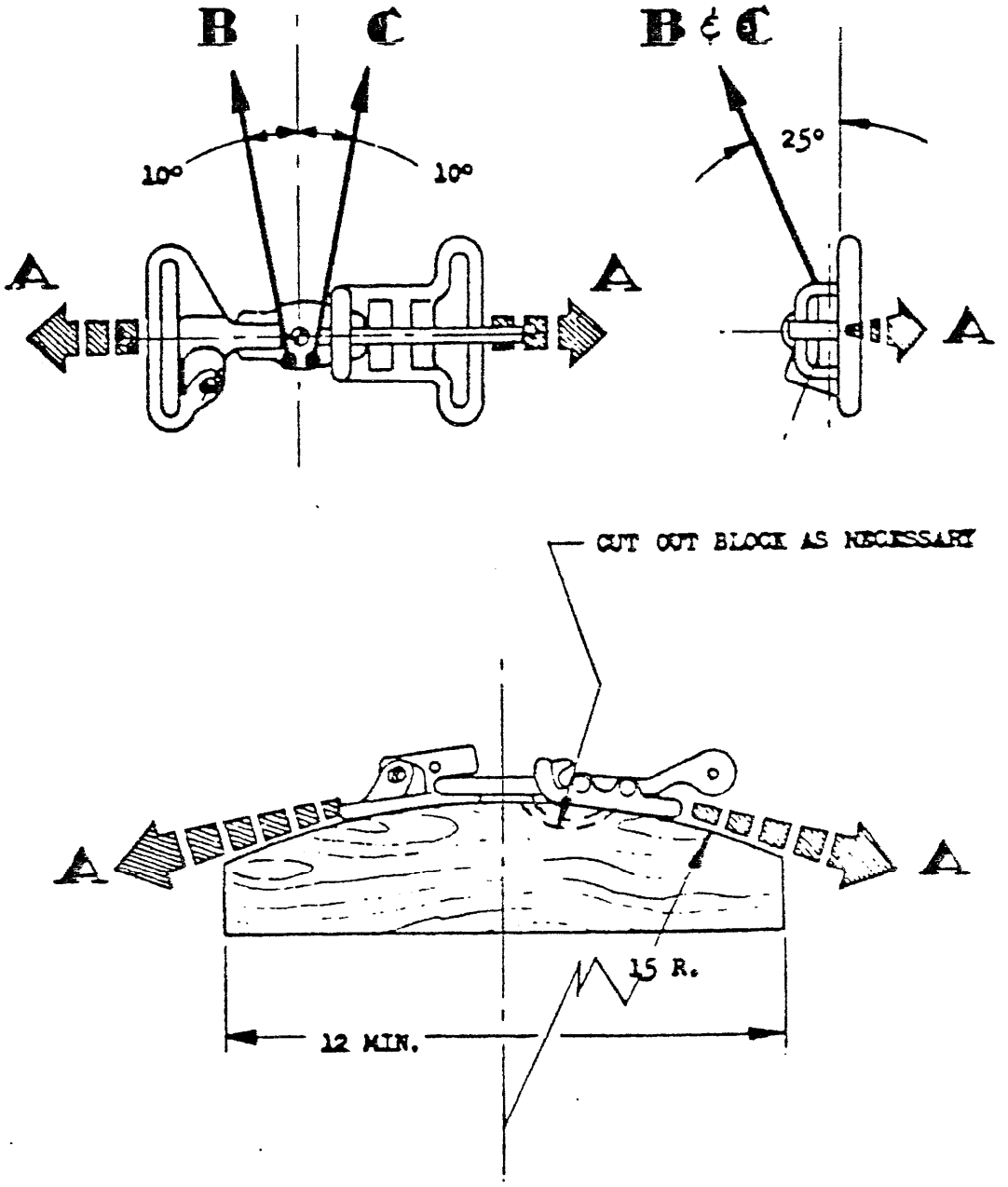
4.4.3 High temperature automatic operation. The automatic lap belt assembly with initiator and a 36-inch length of pressure transmission hose (P/N 55B600-36) attached shall be subjected to a temperature of +160°F (+71°C), for a period of at least three hours. While still at this Temperature, the following loads shall be applied to the belt as shown in figure 1: Load A, 1000 pounds; loads B and C, 0 pounds. The initiator shall then be actuated. The connecting link shall readily separate from the release.

4.4.4 Low temperature automatic operation. The automatic lap belt assembly with initiator and a 48-inch length of pressure Transmission (P/N 55B600-48) attached shall be subjected to a temperature of -65°F (-55°C) for a period of at least 4 hours. While still at this temperature, the following loads shall be applied to the belt as shown in figure 1: Load A, 1000 pounds; loads B and C, 0 pounds. The initiator shall then be actuated. The connecting link shall readily separate from the release.

4.4.5 Ultimate load. The release sssembly shall be mounted on a suitable wooden block, representing the curvature of the anatomy as shown in figure 1, and installed on a suitable static load testing machine. The release shall then be subjected to the following ultimate loads simultaneously: Load A, 2880 pounds; loads B and C, 1800 pounds. There shall be no structural nor mechanical failure of the release. The loads on the release shall then be reduced to the following: Load A_s 100 pounds; loads B and C, 50 pounds. The release shall then be operated automatically. Satisfactory release shall result.

4.4.6 Functional test. After the 2000-pound static load has been applied to each belt assembly, as required by the belt assembly drawings, the AN520-0-4 screw shall be removed and the belt assembly placed in a vertical position with the manual release hanging below the automatic release. Fifty to one hundred psi air pressure of dry filtered air or nitrogen shall be applied to the AN823-3 elbow fitting of the automatic release. The link and lower half of the belt assembly shall fall from the automatic release. Any release which requires more than 100 psi to actuate the piston shall be rejected.

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Dimensions in inches.

Figure 1. Load Diagram

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4.5 First article tests

* 4.5.1 First article test samples. The first article test samples shall consist of models representative of the production equipment. They shall be tested at a laboratory designated by the procuring activity or, when so stated in the contract, at the contractor's plant under the supervision of the procuring activity.

* 4.5.2 First article tests. The first article tests of releases shall consist of all the tests specified in 4.4, Test Methods.

* 5. PACKAGING

* 5.1 Preservation and packaging. Preservation and packaging shall be Level A or C as specified (see 6.2).

* 5.1.1 Level A. Automatic release shall be preserved in accordance with MIL-P-116, Method I.

* 5.1.2 Level C. Automatic release shall be preserved and packaged in accordance with the Manufacturers commercial practice.

* 5.2 Packing. Packing shall be level A, B or C as specified (See 6.2).

* 5.2.1 Level A. Automatic release preserved and packaged to meet 5.1.1 shall be packed in a container conforming to PPP-E-601, Overseas type. The closure of shipping container shall be in accordance with the appendix of the shipping container specification.

* 5.2.2 Level B. Automatic release preserved and packaged as specified in 5.1.1 shall be packed in a container conforming to PPP-B-636, Class Domestic. The closure of shipping container shall be in accordance with the appendix of the shipping container specification.

* 5.2.3 Level C. Each Automatic release preserved and packaged as specified in 5.1.1 shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with the Uniform Freight Classification Rules or regulations of other carriers as applicable to the mode of transportation,

* 5.3 Physical protection. The Automatic release shall be cushioned, anchored, blocked and braced in accordance with MIL-STD-1186. The Freefall Drop Test, Edgewise Drop Test and Cornerwise Drop Test shall be in accordance with Appendix A of MIL-STD-1186. The tests are not required when Level C packing is specified.

* 5.4 Marking. In addition to any special marking required by the contract or order (see 6.2) interior packages and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The automatic release covered by this specification is compatible with the F-1A automatic parachute ripcord release anchor (P/N 55A751) and is intended for use in all USAF aircraft equipped with ejection seats in accordance with Specification MIL-S-6551 and MIL-S-9479. (Reference Drawing 55D599.)

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6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Quantity of Government-furnished property (see 3.3).

c. Applicable levels of preservation and packaging and packing.

6.3 Reconditioning of initiators. After firing, the delay initiator will be disassembled, cleaned, and reloaded by qualified armament, personnel at AF depot level. The procuring activity will take the necessary action to coordinate arrangements for disassembly, cleaning, and reloading of these devices with the cognizant depot. Upon completion of the initial reloading of the initiator, the body and cap should be painted solid red, a narrow black bank painted around the body immediately below the cap, and marked “for ground test only." A second narrow black band should be added on the device after the second reloading. The proper nomenclature should be affixed to the device in accordance with the cartridge installed. These devices should not be fired more than three times, which permits only two reloading on each unit. It must be noted that the components of each device should not be interchanged with those of another unit during disassembly, reloading, and assembly. Upon completion of the third firing of any initiator by the contractor (firing of a unit painted solid red with two black bands), the contractor should dispose of these units as per instructions from the procuring activity.

6.4 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and reiationship to the last previous issue.

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